Module Two — Assignment 2

Electricity Consumption: Preprocessing and Regression

Total: 20 points

Description

In this scenario, you are provided data on the electricity consumption in a town. The data is organized into 7 columns:

- Date Time
- Temperature
- Humidity
- Wind Speed
- General diffuse flows
- Diffuse flows
- Power consumption

Task

Download the data: **M2 Electricity_Consumption-1.csv**

Steps

Conduct data preprocessing. This includes:

- Execute data preprocessing — including verification and handling of repeated rows, addressing missing values, converting categorical variables to numerical variables, and applying feature scaling and feature selection if necessary.
- Analyze and visualize the multicollinearity in the dataset (find correlations between the variables).
- Implement **simple linear regression** to predict the power consumption of the town. Provide results, the accuracy score of the prediction, and a visualization.
- Implement **polynomial linear regression**. Provide results, the accuracy score of the prediction, and a visualization.
- Implement **multiple linear regression**. Provide results, the accuracy score of the prediction, and a visualization.
- Compare and explain the results of the regression models in your own words.

Criteria for Success

The attached rubric indicates how the assignment will be graded.

Note: For simple linear regression, identify the two "best" features that exhibit a linear relationship with the target variable. Apply simple linear regression separately to each feature and compare the scores obtained.

Note: Data preprocessing is mandatory for all assignments. Each assignment should involve the application of all necessary methods for data preprocessing. Neglecting this step will result in a reduction of marks.

Submission Requirements

Students are required to submit two deliverables.

Deliverable One: PDF File. The PDF file should contain the following elements for each question:

- **Screenshots of running code** — screenshots that show the code execution for each question, demonstrating that the code is functional and producing the expected results.
- **Outputs and visualization plots** — all the outputs and visualization plots requested in the questions, properly labelled and annotated for clarity.
- **Explanations and interpretations** — detailed explanations and interpretations of the results obtained from the code, discussing the significance of the outputs and visualizations in relation to the problem's context.
- **Resource citations** — whenever external resources (websites, papers, books, etc.) are used to solve the problems, provide proper citations. This ensures credibility and gives credit to the original sources.
- **Brief report (2–3 pages)** — if a question specifically asks for a report, include a concise report summarizing your findings, methodologies and conclusions. The report should be clear, concise and well organized.
- **Graph and visualization explanations** — avoid one-liners when explaining graphs and visualizations. Provide thorough descriptions that demonstrate a clear understanding of the information presented.

Deliverable Two: Script File (working code). The script file should contain:

- **Functionality** — the code should be free from errors and should run successfully for all the questions. Test your code thoroughly before submitting.
- **Organization** — the script file should be organized logically, with clear comments and explanations for each section of the code. Proper indentation and code formatting are essential for readability.